

Part D

Report of Recommendations: Fraud Detection Model Selection for STADIOalot

This report presents recommendations to STADIOalot based on the results obtained in Part C, where two candidate fraud-detection models were trained and evaluated on the full public benchmark dataset described in Part A of SS2: a Random Forest classifier ("Model 1") and a hybrid Isolation Forest + Neural Network classifier ("Model 2"). It addresses which model is recommended, how that model can be further improved, how it must be adapted to work with the data STADIOalot committed to providing in the SS1 Data Request, and how these findings relate to the wider literature reviewed in Part A.

1. Which model is best between Model 1 and Model 2?

Model 2 (the hybrid Isolation Forest + Neural Network) is better than Model 1 (Random Forest) for production use. The full results are summarised below:

Metric	Model 1 (Random Forest)	Model 2 (Isolation Forest + Neural Network)
Precision	0.2607	0.3538
Recall	0.6718	0.6918
F1	0.3757	0.4681
ROC-AUC	0.9829	0.9723
PR-AUC	0.3352	0.6149
Precision at F1-optimal threshold	0.3290 (thresh. 0.589)	0.7502 (thresh. 0.886)
Recall at F1-optimal threshold	0.5366	0.5193

Model 2's PR-AUC (0.615) is nearly double Model 1's (0.335), and this gap is confirmed as both large and statistically significant by two independent tests: McNemar's test on the two models' hard classification decisions ($\chi^2 = 422.18$, $p = 8.18 \times 10^{-94}$; Model 2 correctly classifies 3,087 transactions that Model 1 gets wrong, against only 1,669 the other way), and a paired bootstrap test on the PR-AUC difference itself (observed difference -0.280 , 95% CI $[-0.298, -0.260]$, meaning the interval never crosses zero).

Model 1 does have a marginally higher ROC-AUC (0.983 vs 0.972), and this difference is also statistically significant (paired bootstrap 95% CI $[0.008, 0.014]$), but ROC-AUC is the less trustworthy metric on a dataset this imbalanced (fraud is only 0.39% of the test set). ROC-AUC rewards a model for correctly ranking the very large legitimate class, which can mask weak performance on the rare fraud class specifically; PR-AUC, precision/recall, and McNemar's test all focus on that rare class directly, and all three unanimously favour Model 2.

The practical difference this makes for a fraud-review team is substantial: at each model's own optimal operating threshold, Model 2 achieves 75% precision at 52% recall (only 371 false alarms across the entire 555,719-row test set), compared to Model 1's 33% precision at 54% recall (2,347 false alarms). Model 2 would require reviewers to investigate roughly a third as many false alarms to catch a comparable number of genuine fraud cases.

One nuance worth noting for future iterations: McNemar's contingency table shows the two models do not simply agree/disagree at random, each one catches a meaningful number of fraud cases the other misses (1,669 cases only Model 1 gets right; 3,087 only Model 2 gets right). This complementary error pattern suggests neither model alone is capturing the full picture and is the basis for the improvement recommendation in the next section.

2. How can the selected model be improved?

- Ensemble with Model 1 rather than discard it.
Since McNemar's test shows the two models make partially independent errors, a stacked ensemble (a meta-learner combining Model 1's and Model 2's probability outputs, as used in Moradi, Tarif and Homaei, 2025) could plausibly recover some of the 1,669 cases Model 1 catches that Model 2 currently misses, without giving up Model 2's PR-AUC advantage.
- Systematic hyperparameter tuning.
Model 2's architecture (2 hidden layers, 5 training epochs, a fixed 0.5 SMOTE ratio, Isolation Forest contamination left at "auto") was set directly from the replicated literature source rather than tuned for this dataset specifically. A grid or random search over the SMOTE sampling ratio, network depth/width, dropout rate, and Isolation Forest contamination would very likely improve on the current result, this is precisely the systematic tuning that Moradi et al. (2025) found gave the largest measured gains in a comparable fraud-detection ensemble.
- Train longer, with early stopping.
The Neural Network's training loss fell steadily from 1.461 to 0.060 across only 5 epochs with no sign of validation divergence, this suggests the network was still improving when training stopped. Extending training with an early-stopping callback monitoring validation PR-AUC (rather than a fixed epoch count) would let the model train exactly as long as it continues to help, without manually guessing a cut-off.
- Threshold selection tied to business cost, not just F1.
The F1-optimal threshold (0.886) is a good general-purpose choice, but STADIOalot's actual cost of a missed fraud versus the cost of a false alarm (reviewer time, customer friction on incorrectly held orders) is not necessarily symmetric. Once those costs are known, the threshold can be tuned directly against expected cost rather than F1, likely shifting it further toward precision or recall depending on which error is more expensive to STADIOalot.
- Expand the feature set once real data is available.
As detailed in Section 3 below, several feature families that the current model could not use (customer sentiment, complaint history, chargeback history) are exactly the kind of "contextual" features that Al-Hajabed (2026) found to be the single most predictive family in this same architecture - adding them is a concrete, literature-supported next step rather than a speculative one.

3. How will the selected model need to be adapted to work with your requested data from SS1?

Model 2 was trained and validated on a public benchmark dataset (the Shenoy, 2020 "Credit Card Transactions Fraud Detection Dataset") because STADIOalot's own transactional data, as specified in the SS1 Data Request, was not yet available for training. Moving Model 2 onto STADIOalot's real data requires several concrete adaptations:

- Remap geolocation-based features to STADIOalot's actual schema.
The benchmark dataset's strongest "contextual" feature was the distance between the cardholder's and merchant's coordinates. The SS1 Data Request does not include customer/seller geolocation, the closest available substitute would be a mismatch flag between the customer's billing province and a seller's registered province (once Seller Profile fields are defined), or a shipping-distance figure derived from Delivery Records once that table's columns are specified.
- Recompute behavioural-velocity features around Customer id, not card number.
The trailing-24-hour transaction count/spend, and time-since-last-transaction features were computed per credit-card number in the benchmark dataset. In STADIOalot's schema these should instead be grouped by Customer id (and cross-referenced with Transaction ID/Payment method), which is a direct, low-risk substitution of grouping key rather than a redesign.
- Incorporate Chargebacks and Flagged Transactions as an improved label source.
The benchmark dataset provided a single binary fraud label. STADIOalot's request explicitly includes both a Chargebacks table and a Flagged Transactions table with a flag reason; this is richer ground truth than the model has used so far. It is worth testing whether defining "fraud" as (flagged OR charged back) produces a cleaner, more precise label than either signal alone, before retraining.
- Add Comments and Star Ratings as new contextual features.
These have no equivalent in the benchmark dataset at all. A sentiment score derived from order Comments, plus the numeric Star Rating, are natural additions to the same "contextual" feature family that Al-Hajabed (2026) found most predictive, this is a genuine opportunity to extend the model beyond what the literature source could test, not just replicate it.
- Finalise the still-undefined data sources before they can be engineered into features.
The Marketplace Data section (Seller profiles, Product listings, Delivery records, Customer complaints) and part of the Customer Data section (Demographics, Search history, Abandoned carts) in the current SS1 Data Request have no columns specified yet. Until those are defined, Model 2 can only be trained on the Transactional, Payments, Chargebacks, and Flagged Transactions tables, the marketplace-seller risk-grouping technique used in the benchmark model (analogous to Carneiro, 2016's merchant risk-tiering) cannot be reproduced for STADIOalot's actual sellers until Seller Profile fields exist.
- Resolve the data-quality issues already flagged in the SS1 Data Request before any timestamp-based feature engineering.
Several of the temporal and categorical features Model 2 relies on (hour-of-day, day-of-week, 24-hour transaction velocity) require consistent, complete date/time fields, but the current Data Request has inconsistent date formats, and several date fields are missing a companion timestamp. Similarly, the model's risk-tiering by location assumed a single consistent geographic field; the current Data Request mixes South African terminology

with example US-style "state" values and does not confirm all amounts are in ZAR. These should be standardised before retraining, not worked around inside the model.

4. Do your results align with the related literature from Part A?

Yes. Three findings from the literature reviewed in Part A are directly echoed by these results:

- A hybrid anomaly-detection + supervised model outperforms a single supervised model. This is the central finding of Al-Hajabed (2026), who reported their hybrid Isolation Forest + Neural Network improving PR-AUC from 0.878 (Neural Network alone) to 0.900 and precision from 0.44 to 0.78. My comparison, a different single model (Random Forest) against the same hybrid architecture, found the same directional result at a larger scale (PR-AUC 0.335 → 0.615), reinforcing rather than merely repeating their conclusion.
- Random Forest is a strong, interpretable baseline, but not always the best performer. Carneiro (2016) and Ileberi, Sun and Wang (2022) both identified Random Forest as their top-performing single classifier, which is consistent with Model 1's very high ROC-AUC (0.983) here. Our results add a nuance neither of those studies tested: when compared specifically against an anomaly-aware hybrid rather than other single classifiers, Random Forest's advantage does not hold on to the metric (PR-AUC) that matters most for this level of class imbalance.
- Combining resampling with a more sophisticated architecture gives the largest gains. Moradi, Tarif and Homaei (2025) found that SMOTE paired with a stacking ensemble produced the biggest measured improvement of any configuration they tested. My result: SMOTE paired with the Isolation Forest + Neural Network hybrid substantially outperforming SMOTE paired with a single Random Forest, is consistent with that same pattern.
- Unsupervised anomaly detection is a viable ingredient for fraud detection. Dornadula and Geetha (2019) demonstrated Isolation Forest and Local Outlier Factor as credible unsupervised approaches when labelled fraud data is scarce or unreliable. Model 2's use of Isolation Forest as a feature-generating first stage builds directly on that premise, and our results support it: the anomaly features it contributes are a meaningful part of why Model 2 outperforms a purely supervised alternative.

There are also two differences worth acknowledging rather than glossing over:

- The class imbalance in our data is more extreme than in some of the literature. Fraud made up only 0.39–0.58% of our dataset, compared to roughly 3.5% in the IEEE-CIS data Moradi et al. (2025) used. This means my absolute PR-AUC figures are not directly comparable to theirs in magnitude, only the qualitative pattern (hybrid/ensemble beats single model; PR-AUC is the metric to trust over ROC-AUC) transfers across studies, not the specific numbers.
- These results were obtained on a public benchmark dataset, not STADIOalot's own data. Al-Hajabed (2026), whose architecture Model 2 replicates, used this same benchmark dataset, so that specific comparison is on equal footing. Carneiro (2016), by contrast, used a real e-tailer's proprietary data; meaning that study's numbers reflect real marketplace conditions in a way ours currently does not. Section 3 above sets out exactly what would need to change before these findings can be confirmed on STADIOalot's own transactions.

References

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